

# TrES-3b

R-1581

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-3\_201015 · created 2026-08-01T23:55:50+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2459137.6658 +/- 0.0015 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.077 +/- 0.053                 |
| Transit depth (Rp/Rs)^2                           | 0.6 +/- 0.82 [%]                |
| Orbital Inclination (inc)                         | 81.88 +/- 2.53                  |
| Airmass coefficient 1 (a1)                        | 1.228 +/- 0.021                 |
| Airmass coefficient 2 (a2)                        | -0.1212 +/- 0.0043              |
| Scatter in the residuals of the lightcurve fit is | 1.67 %                          |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 4.35                            |
| Optimal Annulus                                   | 10.19                           |
| Transit Duration (day)                            | 0.04 +/- 0.027                  |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                            |
|---------------------------|--------------------------------------------|
| Rp/R■ fit vs published    | 0.077 ± 0.053 vs 0.16309 (deviation 1.54σ) |
| Duration fit vs published | 0.04 d vs 0.0567 d (deviation 0.59σ)       |
| Mid-time O–C              | 2.8 min                                    |
| Depth SNR                 | 1.4                                        |
| Residual scatter          | 1.67 %                                     |

### Light curve

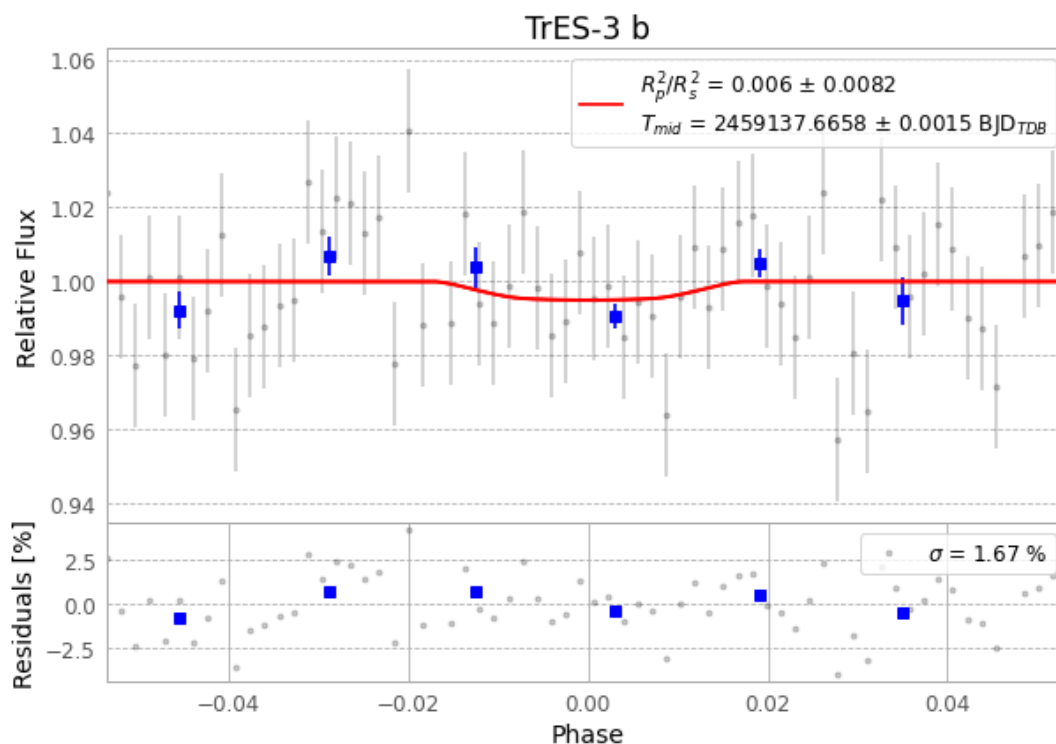


Figure 1 — FinalLightCurve\_TrES-3 b\_2020-10-14.png (SHA-256 da7bbde56297b278...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [270, 260], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 5bf800af15af13a8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

68 FITS frames (45 MB), TRES-3201015021812.FITS → TRES-3201015053912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-201015.log (9cc5afaed1bd...), FinalLightCurve\_TrES-3 b\_2020-10-14.png (da7bbde56297...), FinalLightCurve\_TrES-3 b\_2020-10-14.csv (33102de2807a...)

## Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 23:55Z.